Program Design Quiz 2

(Please use C language to solve the problem)

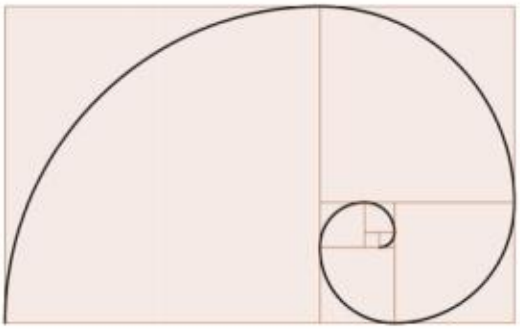
題目敘述：

當代藝術館籌備了一面名為 **Spira Aurea** 的裝置牆。策展人決定以「**黃金比例螺線**」概念佈置：從一個小正方形開始，依序將邊長為費氏數列連續項的正方形**逆時針**向外拼接；在每個方塊內切出四分之一圓，相鄰弧段銜接便形成螺線。完成後，整體外框會是一個長方形，而其長寬比將逼近黃金比例。

你的任務是：給定使用到第 **n** 個方塊，請輸出該外框長方形的「長：寬」。

費氏數列的公式如下：
「 費氏數列：F(0)=0, F(1)=1, F(n)=F(n-1)+F(n-2) {n>=2} 」

如下圖所示的作品，策展人決定使用 8 個方塊，計算出外框長寬比＝21：34。



黃金比例螺線

長寬比和費式數列關係

輸入說明：
第一行為整數 **M**，表示測資組數。
接下來 **M** 行，每行一個整數 n (1 <= n <= 45)，代表使用第 1 到第 **n** 個方塊。

輸出說明：
對每組測資輸出一行，請輸出作品的長寬比 h:w (1 <= h , w <= 2^{31}-1)。
數字與冒號之間沒有空白。

Examples

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Sample Input:

2

6

8

Sample Output:

8:13

21:34

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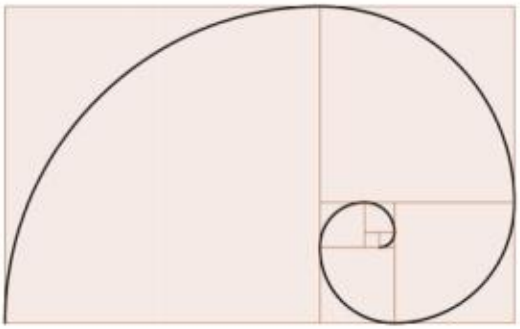
Problem Description：

The Museum of Contemporary Art is preparing an installation wall named **Spira Aurea**. The curator plans to use the concept of the **golden ratio spiral**: starting from a small square, place additional squares outward in a counterclockwise direction, with side lengths equal to consecutive terms of the Fibonacci sequence. In each square, draw a quarter circle; as adjacent arcs connect, they form a spiral. Once completed, the overall outer boundary is a rectangle whose length-to-width ratio approaches the golden ratio.

Your task: given that the first *n* squares are used, output the outer rectangle's aspect ratio **"length:width"**.

The Fibonacci sequence is defined as:
$\mathbf{F(0) = 0, F(1) = 1, F(n) = F(n-1) + F(n-2)}$ for $\mathbf{n \geq 2}$.

As illustrated in the example artwork, when the curator uses **8** squares, the outer rectangle's aspect ratio is **21:34**.



黃金比例螺線

長寬比和費式數列關係

Input：

The first line contains an integer **M**, the number of test cases.
The next **M** lines each contain one integer **n** ($\mathbf{1 \leq n \leq 45}$), indicating that squares **1** through **n** are used.

Output：

For each test case, output one line with the aspect ratio in the form **h:w** (**1 ≤ h, w ≤ 2^31 − 1**).

There must be **no spaces** between the numbers and the colon.

Examples

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Sample Input:

2

6

8

Sample Output:

8:13

21:34